Data Visualization 2

Jameson Roth

```
# ============================================
# Program: retail_data_vis.R
# Author: Jameson Roth
# Date: 2026 10 27
# Description: Produce data visualization 2 - retail counts for different subsets of zips
# ============================================
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------------------------------- tidyverse 2.0
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------------------------------- tidyverse_conflicts
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(rprojroot)
library(RColorBrewer)

setwd(find_root(is_git_root))

pop <- read_csv("0_data/1_clean/base_population.csv") |>
    mutate(cbsa = as.character(cbsa), year = as.character(year)) |>
    select(
        count_totartsentertainment,
        count_coffeeshopscafes,
        count_delis,
        count_fastfood,
        count_dollarstores,
        cbsa_name,
        cbsa,
        year,
        low_income,
        high_income,
        zcta,
        zipcode,
        zip_rank
    ) |>
    # TODO: move this to retail cleaning code
    pivot_longer(
        starts_with("count"),
```

```
        values_to = "count",
        names_to = "category",
        names_prefix = "count_"
    )
```

```
## Rows: 2241 Columns: 35
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (3): zipcode, zcta, cbsa_name
## dbl (28): statefips, year, cbsa, zip_median, zip_rank, annual_avg_zhvi, medi...
## lgl  (4): gentrifiable, low_income, high_income, gentrifying
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# TODO: control for cbsa-year population size or number of total businesses in cbsa-year
↪  to make this more accurate

grouped_lo <- pop |>
    filter(low_income) |>
    group_by(cbsa_name, year, category) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

```
summary(grouped_lo)
```

```
##      cbsa_name             year         category      count
##  Length   :30   Length   :30   Length   :30   Min.   : 0.4828
##  N.unique : 6   N.unique : 1   N.unique : 5   1st Qu.: 2.7831
##  N.blank  : 0   N.blank  : 0   N.blank  : 0   Median : 4.3460
##  Min.nchar: 6   Min.nchar: 4   Min.nchar: 5   Mean   : 4.7322
##  Max.nchar:15   Max.nchar: 4   Max.nchar:20   3rd Qu.: 5.9084
##                                               Max.   :12.3276
```

```
grouped_hi <- pop |>
    filter(high_income) |>
    group_by(cbsa_name, year, category) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

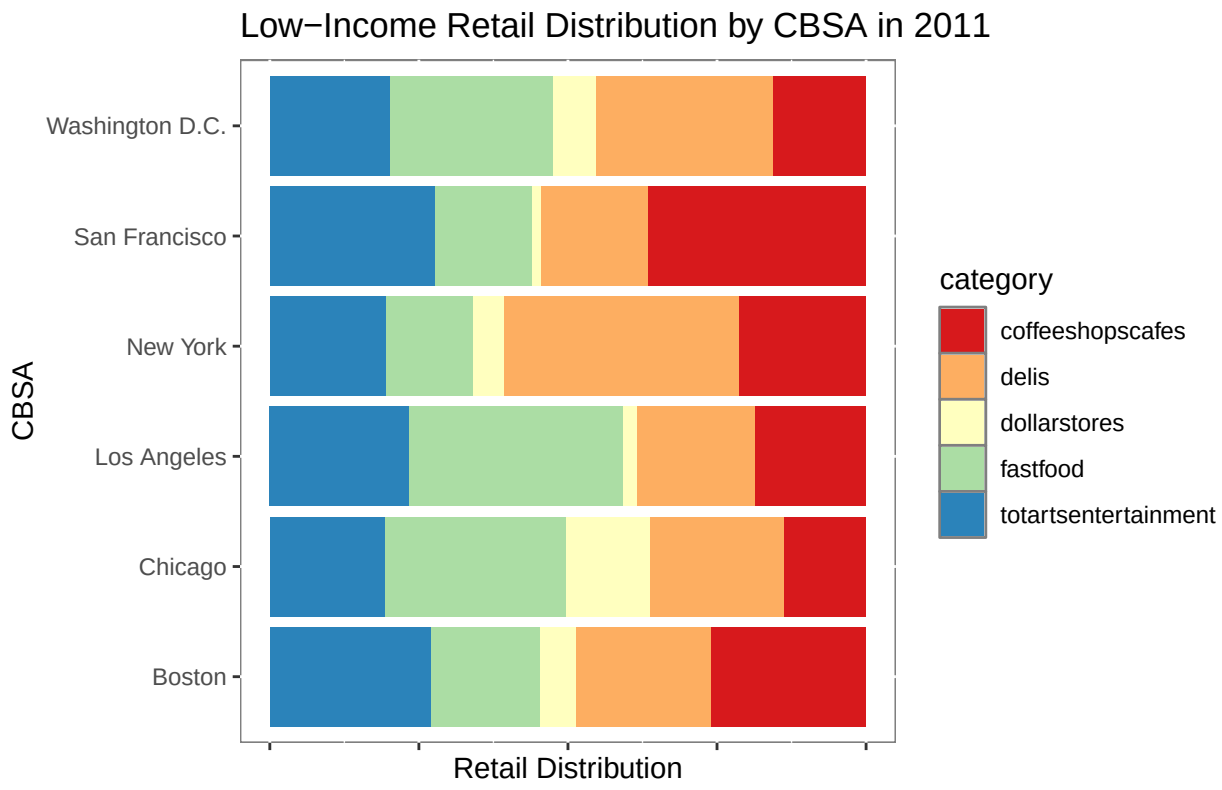
```
summary(grouped_hi)
```

```
##      cbsa_name          year          category      count
##  Length   :30   Length   :30   Length   :30   Min.   :0.000
##  N.unique : 6   N.unique : 1   N.unique : 5   1st Qu.:1.168
##  N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :2.962
##  Min.nchar: 6   Min.nchar: 4   Min.nchar: 5   Mean   :2.876
##  Max.nchar:15   Max.nchar: 4   Max.nchar:20   3rd Qu.:4.482
##                                               Max.   :6.429
```

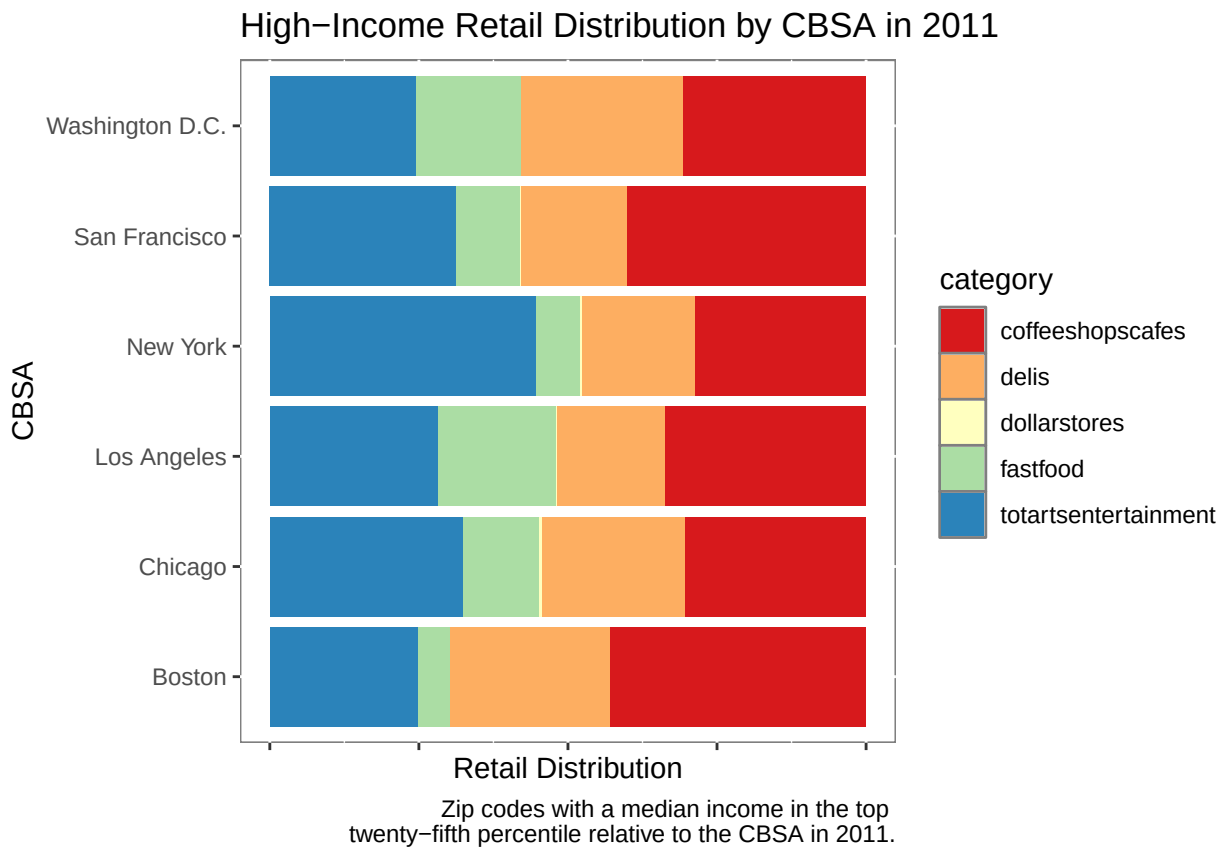
```
grouped_med <- pop |>
    filter(zip_rank > 50, zip_rank < 75) |>
    group_by(cbsa_name, year, category) |>
    summarize(count = mean(count)) |>
    ungroup()
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by cbsa_name, year, and category.
## i Output is grouped by cbsa_name and year.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(cbsa_name, year, category))` for per-operation grouping (`?dplyr::dplyr_by`)
##   instead.
```

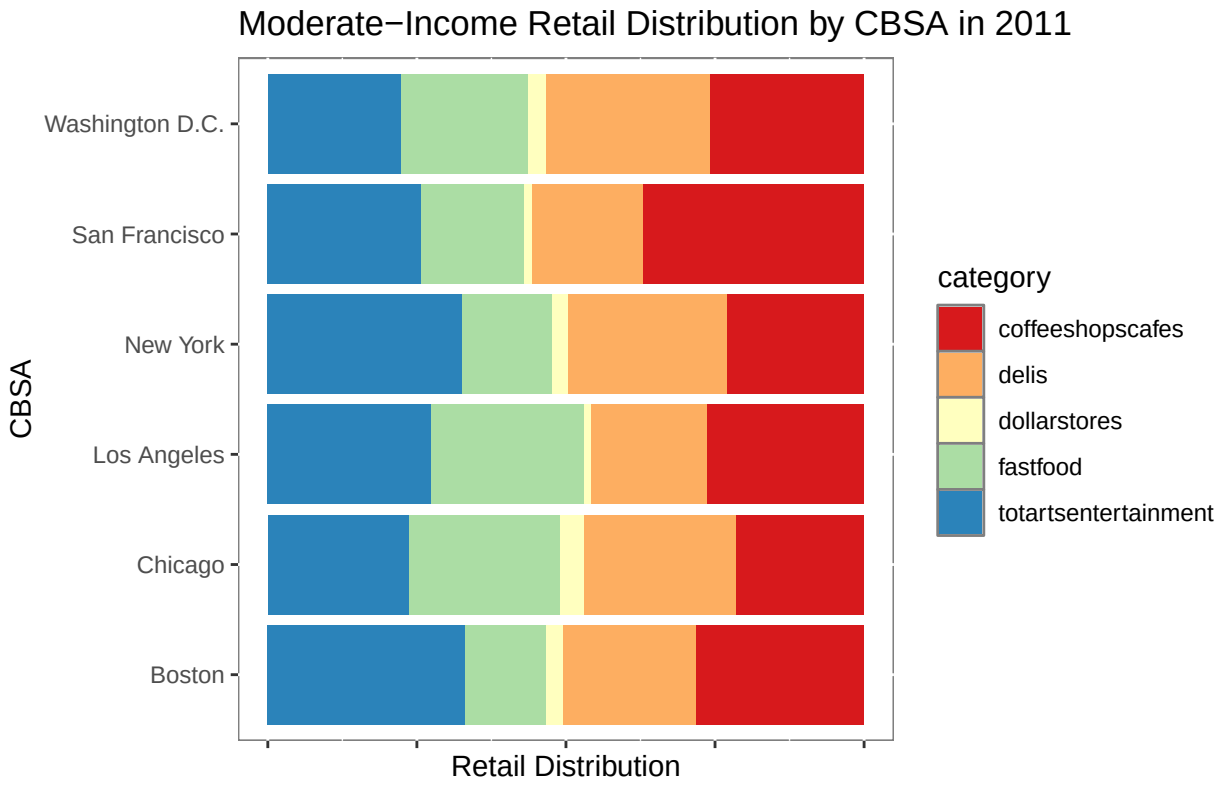
```
ggplot(
    grouped_lo,
    aes(y = cbsa_name, x = count, fill = category)
) +
    geom_bar(stat = "identity", position = "fill") +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "Low-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income in the \n bottom fiftieth percentile
        ↪  relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
    ) +
    scale_fill_brewer(palette = "Spectral")
```




```
ggplot(
    grouped_hi,
    aes(y = cbsa_name, x = count, fill = category)
) +
    geom_bar(stat = "identity", position = "fill") +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "High-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income in the top \n twenty-fifth percentile
        ↪  relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
    ) +
    scale_fill_brewer(palette = "Spectral")
```




```
ggplot(
    grouped_med,
    aes(y = cbsa_name, x = count, fill = category)
) +
    geom_bar(stat = "identity", position = "fill") +
    xlab("Retail Distribution") +
    ylab("CBSA") +
    labs(
        title = "Moderate-Income Retail Distribution by CBSA in 2011",
        caption = "Zip codes with a median income below the top twenty-fifth \n
        ↪  percentile and above the fiftieth percentile relative to the CBSA in 2011."
    ) +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank(),
    ) +
    scale_fill_brewer(palette = "Spectral")
```



Zip codes with a median income below the top twenty−fifth percentile and above the fiftieth percentile relative to the CBSA in 2011.